Mock Questions - IE2024

Probability and Statistics

Important Notice to Students

These mock questions are provided strictly for practice purposes only. They do not provide any hints, predictions, or indication about the questions or structure of the final examination. Students must study all lecture notes and the complete syllabus for the final examination. Multiple-choice questions (MCQs) are not included in this practice document; MCQs in the final examination will also cover the full syllabus.

Instructions: Answer using clear statistical reasoning. Show relevant formulas, complete calculation steps, and interpretations. Each question has three subquestions.

Question 01

A university has 4,000 undergraduate students divided into three academic programs. The university wants to select 200 students to investigate satisfaction with online learning. The research team decides to use proportionate stratified random sampling.

Academic program	Number of students
Computer Science	2,000
Information Technology	1,200
Data Science	800
Total	4,000

a) Explain why stratified random sampling is suitable for this study, considering the differences in student numbers across the three programs.

b) Calculate the number of students that should be selected from each academic program for a total sample of 200 students.

c) Analyze how sample representativeness could be affected if researchers selected 100 students from Computer Science, 50 from Information Technology, and 50 from Data Science instead.

Question 02

A software company investigates the time required to complete a programming task. Seven developers record completion times in minutes: 18, 20, 21, 22, 24, 25, 80. The project manager wants to report a typical completion time, but one developer took much longer than the others.

a) Calculate the mean, median, and mode of the completion times, stating clearly if no mode exists.

b) Calculate the range and interquartile range (IQR) using the median-of-halves method, and identify any outliers using the 1.5 × IQR rule.

c) Analyze how the unusually large completion time affects the mean and median, and explain which measure better represents typical completion time.

Question 03

A hospital compares the daily number of patients treated by two medical units over five days. The hospital director wants to identify which unit has more consistent daily patient numbers.

Day	Unit A	Unit B
1	18	10
2	20	15
3	22	22
4	24	29
5	26	36

a) Explain the difference between variance and standard deviation, considering their usefulness in measuring variability.

b) Calculate the sample variance and sample standard deviation of daily patient numbers for both units.

c) Analyze the calculated results and identify which unit has more consistent patient numbers, considering both average and variability.

Question 04

A retail company studies whether customers who use its mobile application are more likely to purchase products during a promotional campaign. Information was collected from 500 customers. A customer is randomly selected from the dataset.

Customer category	Purchased	Did not purchase	Total
App users	180	120	300
Non-app users	80	120	200
Total	260	240	500

a) Explain the difference between independent events and mutually exclusive events using the events “customer uses the app” and “customer makes a purchase.”

b) Calculate the probability that the selected customer uses the app or makes a purchase.

c) Calculate the probability that a customer makes a purchase given that the customer uses the app, and analyze whether app usage and purchasing are statistically independent.

Question 05

An airport introduces an automated baggage scanning system. Historical records show that 4% of all bags contain a prohibited item. The system correctly detects 95% of bags containing prohibited items but incorrectly flags 8% of bags that do not contain prohibited items.

a) Explain the difference between prior probability and conditional probability in this baggage scanning system.

b) Using Bayes’ theorem, calculate the probability that a bag actually contains a prohibited item when the scanner produces a positive result.

c) Analyze why a positive scanning result does not necessarily mean that a bag contains a prohibited item, considering the frequency of prohibited items and the false-positive rate.

Question 06

An electronics manufacturer produces computer processors. Historical records show that 5% are defective. A quality control officer randomly selects 12 processors from a large production batch. Assume independent defect statuses and constant defect probability.

a) Explain why the Binomial distribution is suitable for modelling the number of defective processors.

b) Calculate the probability that exactly two processors are defective among the 12 selected.

c) Calculate the expected number and variance of defective processors, and analyze what these values indicate about production quality.

Question 07

A university IT help desk receives an average of six technical support requests per hour. The number of requests follows a Poisson distribution with a constant arrival rate.

a) Explain the main characteristics of the Poisson distribution and why it suits technical support requests.

b) Calculate the probability that exactly four requests are received in one hour.

c) Calculate the probability that more than two requests are received in 30 minutes, and analyze how this could help staffing.

Question 08

A food delivery company records delivery times. Historical data suggest that times are approximately normally distributed with a mean of 35 minutes and a standard deviation of 8 minutes. The company promises delivery within 45 minutes.

a) Explain the importance of mean and standard deviation when describing a normal distribution of delivery times.

b) Calculate the probability that a randomly selected order takes more than 45 minutes.

c) Calculate the delivery time corresponding to the 90th percentile using $z_{0.90} = 1.282$, and analyze what this means for the delivery promise.

Question 09

A telecommunications company studies waiting time between consecutive network failures. Historical records suggest an exponential distribution with an average of 10 days between failures.

a) Explain the memoryless property of the exponential distribution in the context of network failures.

b) Calculate the probability that the next network failure occurs within five days.

c) Calculate the probability that the next failure occurs within the next three days given that the system has operated seven days without failure, and analyze the result using the memoryless property.

Question 10

An online examination platform records assessment completion times. The population is positively skewed, with mean 60 minutes and standard deviation 18 minutes. A random sample of 81 students is selected.

a) Explain the Central Limit Theorem and why the sampling distribution of the sample mean can be approximately normal even when the population is skewed.

b) Calculate the standard error for samples of 81 and the probability that the sample mean exceeds 63 minutes.

c) Analyze how increasing the sample size from 81 to 324 affects standard error and precision.

Question 11

A beverage manufacturer estimates the average amount of liquid filled into bottles. A random sample of 25 bottles from a normal population has mean 502 ml and sample standard deviation 10 ml. Population standard deviation is unknown. Use $t_{0.025,24} = 2.064$.

a) Explain why the Student's t-distribution is appropriate for the confidence interval.

b) Calculate the 95% confidence interval for the population mean filling volume.

c) Analyze whether the interval supports a claim that the average filling volume is 500 ml and explain the correct interpretation of a 95% confidence interval.

Question 12

A mobile phone manufacturer claims mean battery life is 20 hours. A testing organization randomly selects 36 phones with mean 18.8 hours. The known population standard deviation is 3.6 hours. Assume independent observations and an appropriate Z-test. Use a 5% significance level and left-tailed critical value −1.645.

a) State the null and alternative hypotheses and explain the meaning of a Type I error.

b) Calculate the test statistic and perform the hypothesis test at 5% significance.

c) Analyze the statistical conclusion and whether there is sufficient evidence to reject the manufacturer's claim.

Question 13

A public transport authority investigates whether preferred transport method is associated with employment status. A survey of 200 passengers provides the following table. Use a 5% significance level and $\chi^2_{0.05,2} = 5.991$.

Employment status	Bus	Train	Taxi	Total

Employed	50	40	30	120
Unemployed	40	20	20	80
Total	90	60	50	200

a) Explain why the Chi-Square Test of Independence suits these categorical variables.

b) Calculate the expected frequencies and Chi-Square test statistic.

c) Analyze the result and determine whether there is sufficient statistical evidence of an association.

Question 14

A property development company predicts apartment selling price using floor area. The company collects data from five recently sold apartments and chooses Simple Linear Regression with area as the independent variable and price as the dependent variable.

Floor area (m^2)	Selling price (Rs. million)
50	12
60	14
70	15
80	18
90	21

a) Explain the slope and intercept in this apartment-price regression model.

b) Using least squares, calculate the estimated regression coefficients and write the fitted regression equation.

c) Predict the price for 75 m^2, calculate the residual for the 80 m^2 apartment, and analyze what this residual indicates.

Question 15

A hotel studies quarterly room revenue. Seasonal fluctuations become larger as overall revenue grows. The team estimates the following components for a quarter and wants to use a multiplicative time series model.

Component	Estimated value
Trend (T)	Rs. 8 million
Seasonal index (S)	1.25
Cyclical index (C)	0.95
Irregular index (I)	1.04

a) Explain additive versus multiplicative time series models, considering seasonal variations as trend increases.

b) Using the multiplicative time series model, calculate estimated quarterly revenue.

c) Analyze how estimated revenue changes if the seasonal index decreases from 1.25 to 0.90 while other components stay unchanged, and explain the practical meaning.